

Software Requirements Specification (SRS): Foodies Web App

1. Introduction

1.1 Purpose

This document specifies the software requirements for the Foodies Web App. It defines the system's intended behavior, and functional features, based on customer Requirements Specification (CRS), and System Integration Questions (SIQs).

1.2 Project Scope

Foodies is a PC-based web platform designed to connect users with nearby restaurants for food discovery and ordering. The system features user authentication, a 10km-radius proximity search, cash-on-delivery ordering, and an integrated loyalty credit system.

2. Constraints

This section outlines the boundaries and technical limitations of the system.

- **2.1 Technical Constraints**

Platform: The application is strictly a Web-based system optimized for PC/Desktop browsers.

Unique Identifiers: The system must use the user's Email Address as the unique ID. No auto-generated numeric IDs shall be used.

Payment Gateway: The system is restricted to Cash on Delivery (COD) only. No online payment integrations are supported at this stage.

- **2.2 Business & Operational Constraints**

Service Scope: The platform is limited to standard food ordering. Wedding catering services (or events exceeding 100 persons) are explicitly out of scope.

Communication: Ordering is conducted exclusively through the web interface. No hotlines or phone-based ordering support is available.

Geographic Limit: Restaurant visibility and ordering are restricted to a 10km radius from the user's location.

3. Operating Environment

This section describes the environment in which the software will operate, including the hardware platform, Software, and Network.

- **Software:** Accessible via any modern web browser (Chrome, Firefox, Edge, Safari).
 - **Hardware:** PC and Desktop use.
 - **Network:** Requires a stable internet connection for real-time location and order processing.
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4. Informative Requirements

4.1 Restaurant Eligibility & Scope

- **RES-IR-01 Restaurant Eligibility & Scope:**
The system supports only food-ordering services. To be listed on the platform, a restaurant must meet the following criteria:
 - **Allowed:** Restaurants that serve food, with or without drinks.
 - **Prohibited:** Restaurants that serve only drinks are excluded.

4.2 Balance Usage

- **BU-IR-01 Balance Usage:** Balance only can be used to buy inside the app.

4.3 Payment Methods

- **PAY-IR-01 Payment Method:**

The website only supports one method of payment. The user can ONLY pay using "Cash on Delivery".

4.4 Ordering

- **ORD-FR-01 Ordering Method:**

The user can ONLY order the food from the website, not a hotline.

5.11 Administrative Features

- **ADM-FR-04 Admin Scope:**

The Admin role is a global position with full access. They have the permission to view and manage the data, schedules, and promotions for every restaurant registered on the platform.

5. Functional Requirements

5.1 User Account Management

- **UA-FR-01 Registration:**

Users sign up using email and a password and then are redirected to the login page.

- **UA-FR-02 Password Constraints:**

8–64 characters, one capital, one small, one special character, and one number, ALL the constraints must be met, e.x. the user has to respect the 5 constraints in order to create a valid password.

- **UA-FR-03 Email Validation:**

When the user provides invalid email, e.g. no “@” or invalid domain name, the user sees an error message saying “Invalid email”.

- **UA-FR-04 Unique Identification:**
The user's email address serves as their unique ID in the system.
 - **UA-FR-05 Email Field Validation:**
If the field is left empty an error message appears to the user saying "Email can't be empty".
 - **UA-FR-06 Password Field Validation:**
If the field is left empty an error message appears to the user saying "Password field can't be empty".
 - **UA-FR-07 Error Message Placement:**
All the error messages are inlined (under the field).
 - **UA-FR-08 Real-Time Validation:**
All the error messages appear in real time to the user, e.g. the user left a field empty the error message appears to the user at run time with the error message.
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5.2 User Login Management

- **LG-FR-01 Login Authentication:**
Registered users can access their accounts by providing their registered email and password.
 - **LG-FR-02 Login Error – Unregistered Email:**
If the entered email is not registered in the system, a general error message appears to the user saying "Check email and password again".
 - **LG-FR-03 Login Error – Wrongly Typed Email:**
If the entered email is registered but wrongly typed in, a general error message appears to the user saying "Check email and password again".
 - **LG-FR-04 Login Error – Incorrect Password:**
If the entered password is incorrect, a general error message appears to the user saying "Check email and password again".
 - **LG-FR-05 Login Error Timing:**
All login error messages appear ONLY after the user clicks submit, not in real-time.
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5.3 Home Page

- **HP-FR-01 Home Redirection:**
After successful login, the user will be redirected automatically to the Home Page.
 - **HP-FR-02 Nearby Restaurant Listing:**
The Home Page displays a list of all restaurants located within a 10 km radius from the user's saved location.
 - **HP-FR-03 Empty Nearby State:**
If no restaurants are available within the defined 10 km radius, the system displays a message saying: "No restaurants available nearby".
 - **HP-FR-04 Home Navigation:** The home page includes a "**Search bar**," "**Restaurants (in 10km radius)**", "**Offers tab**," and "**My Account tab**." Clicking on each tab redirects the user to the corresponding page or menu.
 - **HP-FR-05 Dynamic Cart Visibility:** The "**Cart**" tab is hidden by default on the Home Page navigation. It only becomes visible and accessible once the user has added at least one (1) item to their cart. If the cart is cleared or emptied, the tab disappears from the Home Page again.
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5.4 Restaurant

- **RES-FR-01 Restaurant Service Limitation:**
The system permits users to place bulk orders, limited to a maximum of 100 total items per single order.
 - **RES-FR-01.1 Combo Item Definition:**
For the purpose of calculating the 100-item maximum limit, a bundled product or "combo meal" (e.g., a meal containing a burger, fries, and a drink under a single price) is counted by the system as **one (1) item**.
 - **RES-FR-02 Restaurant Information:**
For each restaurant the delivery time, delivery price and rating are defined in the restaurant's own page, so the user can select the right restaurant accordingly.
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5.5 Ordering

- **ORD-FR-01 Single Restaurant Rule:**

The user can ONLY order from one restaurant at a time.

- **ORD-FR-01.1 Multiple Restaurant Error:**

If the user tries to order from multiple restaurants at a time an error message "You can order from only one restaurant at a time" appears.

- **ORD-FR-01.2 Bulk Limit Error:**

If a user attempts to add more than 100 items to their cart, an error message appears to the user that says: "Maximum limit of 100 items per order is reached."

- **ORD-FR-02 Add to Cart Interaction:**

After the user selects the quantity and clicks on "Add Item," a pop-up appears to the user with two options: "Continue Ordering" and "View Cart." Clicking "Continue Ordering" allows the user to keep browsing the menu, while clicking "View Cart" redirects the user to the cart page.

- **ORD-FR-03 Cart Management:**

Inside the "View Cart" page, the user has the ability to manage their selection by increasing the quantity of an item, decreasing the quantity, or completely removing an item from the cart.

5.6 Offering & Promotions

- **OFF-FR-01 Offer Navigation:**

Users can find the offer screen from the home screen where all the offers are available for all restaurants, or from the restaurant menu of a specified restaurant then it will ONLY display the offers of this specific restaurant.

- **OFF-FR-02 Offer Expiration:**

The offers will disappear by default after the selected duration for the offer, e.g. the offer expires in 3 days -> after this period of time the offer will disappear automatically.

- **OFF-FR-03 Promotion Transparency:**

On each item on discount, there will be the percentage of the promotion and the before and after price.

5.7 Search

SRCH-FR-01 Default Sorting Logic:

By default, the system must display search results based on Nearest (Distance).

SRCH-FR-01.1 Sorting Criteria (Dropdown):

The system provides a dropdown menu allowing users to override the default sort:

- **Distance:**
 - **Nearest:** Ascending order of distance (Default).
 - **Farthest:** Descending order of distance (useful for checking delivery boundaries).
- **Rating (Rate):**
 - **High to Low:** Descending order of star ratings (e.g., 5.0 → 1.0).
 - **Low to High:** Ascending order of star ratings (useful for testing "hidden" or poorly rated restaurants).

SRCH-FR-01.2 Tie-Breaking Rule:

In the event of a tie where two or more restaurants have the same primary sort value (e.g., identical distances or identical ratings), the system applies a secondary sort **Alphabetically by Name (A-Z)**.

Example: If "Burger Palace" and "Apple Cafe" are both 2.3 km away, "Apple Cafe" will appear first.

SRCH-FR-01.3 Handling New/Unrated Restaurants:

The system treats restaurants with no existing ratings (null) as having a value of 0.0.

- **High to Low:** These restaurants will appear at the bottom of the list.
- **Low to High:** These restaurants will appear at the top of the list.

SRCH-FR-02 Syntax Error Handling:

If the user entered a syntax error, e.g. Borgar instead of Burger, the system handles the syntax error and recommends based on similarity of the typed in name.

SRCH-FR-03 No Matching Restaurant:

If the user entered a restaurant name that is not in the system in the search bar an error message will appear to the user saying “No matching restaurants”.

SRCH-FR-04 Out of Range Message:

If the user is far from the restaurant delivery area, an error message will appear to the user saying "This restaurant is out of range".

5.8 Payment & Checkout

PAY-FR-01 Order Calculation:

Before setting the order, the system calculates and displays the total checkout price as the sum of the food price and the defined delivery price for that specific restaurant.

PAY-FR-02 Checkout Address Management

The checkout page allow users to provide a delivery address through two primary methods:

- **Saved Address Selection (AC-FR-02.1):**
 - If the user has saved an address in their profile, it will be displayed as selectable cards.
 - Selecting a saved address fulfills the requirement for the delivery location.

- The selected address should be visually highlighted (e.g., with a darker border or a selection dot).
- **Manual Address Entry:**
 - If no saved addresses exist, or if the user chooses to enter a new one, specific manual input fields must be provided.
 - **Phone Number** (Required - Must follow **AC-FR-02.2 Phone Number** constraints)
 - **Building Name** (Required)
 - **Apartment Name/Number** (Required)
 - **Floor** (Required)
 - **Street** (Required)
 - **Nearby Landmark** (Optional)
 - **Data Persistence Rule:** Addresses entered manually through this method are strictly for **one-time use**.
 - **Profile Limitation:** The system will not save manually entered addresses to the user's "My Account" tab or profile.
 - **Re-entry Requirement:** Users must re-enter their details for every subsequent order unless they explicitly save an address to their profile beforehand.

PAY-FR-03: Point Redemption Checkbox

During Checkout, the system shall display the user's current **point balance** and its **EGP** value.

- **Rule:** A checkbox is provided to apply the points.
- **Threshold Lock:** The checkbox is only **enabled** if the balance is > 1000 points.
- **Validation:** If checked, the system calculates the equivalent EGP value and **subtracts it from the Order Total** before the final payment is processed.

PAY-FR-04 Payment Confirmation: The system displays the payment method, which is strictly limited to "Cash on Delivery" (COD).

PAY-FR-04.1 (Missing Address Error): If the user attempts to proceed without selecting a saved address or completing the manual entry fields, an error message appears to the user "**Can't place an order without specifying the address**".

PAY-FR-04.2 (Button State): The **"Place order"** button remains disabled, it cannot be clicked until a valid delivery address is provided. The button only activates once all required address information is filled or selected.

5.9 Loyalty & Credit System

LOY-FR-01 Point Accrual Rate:

The system shall automatically calculate loyalty points for every successful order. The accrual rate is set to **0.1 point for every 1 EGP**, e.g. **1000 points = 10 EGP** spent on the order subtotal (excluding delivery fees), these are store credit where the user can use this balance to pay with at the checkout screen.

Users are able to view their conversion progress via a progress bar on the **Loyalty Dashboard** to know how far or close they are to redeem their points.

LOY-FR-02 Point Rounding Rule (Floor):

The system applies a "Floor" rounding logic to all point calculations. Any floating-point results must be truncated to the nearest whole integer.

- *Example:* An order of 85.99 EGP results in 8.599 points, which the system must record as **8 points**.

LOY-FR-03 Point Integrity & Default State:

- **Default Value:** Every new account shall be initialized with a point balance of **0**.
 - **Non-Negativity:** The point balance is a non-negative integer. Under no circumstances (including order cancellations or refunds) shall the balance drop below zero.
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5.10 My Account

AC-FR-01 My Account Navigation: The user can access a "My Account" menu which contains three primary options: "Personal Info," "Admin Panel (For admins only) ," and "Log Out."

AC-FR-02 Personal Information Management:

Clicking the "Personal Info" tab redirects the user to a dedicated page where they can view and update their profile details. This page includes the following tabs and each tab redirects the user to its corresponding page:

- **Name:** Allows the user to update their display name.
- **Password:** Allows the user to update their login credentials.

AC-FR-02.1 Saved Address Details: The "Saved Address" section allows the user to manage their primary delivery location using the following specific fields:

1. **Building Name** (Required)
2. **Apartment Name/Number** (Required)
3. **Floor** (Required)
4. **Street** (Required)
5. **Nearby Landmark** (Optional)

AC-FR-02.1.1 Phone Number: The phone number field **only** accepts numeric digits (inside the Saved addressed page). The field is limited to exactly **11 digits** (Egypt mobile format) entered manually by the user.

- **Length Error:** If the user enters more or fewer than 11 digits, an error message appears: **"Phone number must be exactly 11 digits."**
- **Type Error:** If the user attempts to enter letters or special characters, the system must prevent the input or display: **"Only numbers are allowed."**

AC-FR-04 Account Logout: The "Log Out" button allows the user to securely end their current session. Upon clicking, the user is signed out of the system and redirected to the login page.

AC-FR-05 Admin panel: This section of the "My account" is **only** visible for admins not for regular users. Upon clicking on it, it redirects the user to the "Admin panel" where the admin can manage their features discussed later in **"5.11 Administrative Features"**

5.11 Administrative Features

ADM-FR-01 Restaurant Management:

The Admin has the authority to manage the restaurant list. This includes adding new restaurants, updating existing details, viewing the full directory of all restaurants, and removing them from the system when necessary. The Admin sees a list of all businesses and can click "Edit" to change a name or working hours or "Delete" to take them off the app.

ADM-FR-02: Operational Schedule Enforcement

The system must automatically enforce restaurant operating hours based on the server's clock.

- **Status Validation:** The system continuously compares the current server time against the restaurant's defined open/close windows.
- **Action Restriction:** When the current time falls outside of the operating window (e.g., at 10:01 PM for a 10:00 PM closing), the "Add" button for all items associated with that restaurant enter a **disabled state** and become unclickable.
- **User Notification:** A clear notification message must appear directly above the disabled button stating: **"This restaurant is currently closed and not accepting orders."**

ADM-FR-03 Promotion Control:

The Admin is responsible for managing marketing offers. They can create new promotions for any restaurant and set a specific duration (start and end times) for how long the discount is visible to users. Also they can edit existing promotions by changing the names or their time. This relies on "Start" and "End" dates. When the Admin sets a 24-hour promo, the system runs a timer. Once the timer hits zero, the discount tag is automatically removed from the app without the Admin having to do anything manually.

6. Stakeholders

- **End User (Customer):** Browses, searches, and orders food; manages loyalty points.
- **Admin:** Performs restaurant management, adjusts hours, and manages promotions.